GITAM Deemed to be University

Aerospace Engineering | Course Code: AE301

Aerodynamics & Flight Mechanics - Laboratory Manual & Practical Guide

Syllabus Focus: AE301 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Fundamental Incompressible Potential Flow

" Syllabus Scope: Continuity and Euler equations, Velocity potential, Stream function, Elementary flows S

Unit 2: Incompressible Airfoil & Finite Wing Theory

" Syllabus Scope: Kutta condition, Thin airfoil theory for symmetric and cambered airfoils, Prandtl lifti

AE301: Aerodynamics & Flight Mechanics - Advanced Topics & Assessment

Academic Track: Aerospace Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Compressible Flow & Shock Waves

" Speed of sound, Mach number regimes, 1D isentropic flow through converging-diverging nozzles, Normal and obl

Unit 4: Aircraft Performance & Flight Stability

" Thrust required and power required curves, Rate of climb, Service ceiling, Static longitudinal stability, Ne

Unit 5: Hypersonic Aerodynamics & Viscous Effects

" Newtonian impact theory, Boundary layer transition, Aerodynamic heating, Swept wing benefits, Shock stall

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Surface Pressure Distribution Measurement over a NACA 0012 Airfoil in a Subsonic Wind Tunnel

Lab 2: Lift and Drag Force Measurement using Three-Component Wind Tunnel Balance at Varying Angles of Att

Lab 3: Flow Visualization over an Aircraft Model using Smoke Injection and Wool Tufts

Lab 4: Supersonic Shock Wave Visualization in a Converging-Diverging Nozzle using Schlieren Optics

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* John D. Anderson Jr. - Fundamentals of Aerodynamics McGraw-Hill 6th Ed

* E.L. Houghton, P.W. Carpenter - Aerodynamics for Engineering Students Butterworth-Heinemann

Authorized NoteVault Educational Document " Verified for Registered University Students